

192.204 Logic-based Artificial Intelligence

Project 1 Specification - Description Logics and OWL

The deadline for submitting your solution is on **Wednesday, May 20, 23:55**. Please submit early and do not wait until the deadline.

You can get up to **13 points + 1 optional bonus point** for this project: **9 points** for the OWL implementation, **4 points** for the analysis questions, and **1 point** for the AI usage disclosure.

Submit a ZIP file containing:

- `neighborhoods_XXXXXXXXX.rdf` - Your OWL ontology file
- `answers_XXXXXXXXX.pdf` - Your answers to the analysis questions and the AI usage disclosure

where XXXXXXXXXX are the 8 digits of your student ID.

For questions on the assignment please consult the TUWEL forum. Questions that reveal (part of) your solution should be discussed privately via email to `shqiponja.ahmetaj@tuwien.ac.at`.

1 The Neighborhood Knowledge Base

The objective is to specify a Description Logic Knowledge Base (KB) about terminological knowledge on neighborhoods in the *Web Ontology Language* (OWL) and to implement it using the OWL editor of Protégé 5.6.3.

Before you start designing your ontology we recommend reading the following tutorials to make yourself familiar with Protégé 5.6.3 and ontology engineering.

- <http://protegeproject.github.io/protege/getting-started/>
- More detailed information on Protégé can be found at http://mowl-power.cs.man.ac.uk/protegeowltutorial/resources/ProtegeOWLTutorialP4_v1_3.pdf
- An ontology development tutorial can be found at http://protege.stanford.edu/publications/ontology_development/ontology101.html
- In addition to this online-resources we provide Chapter 8 ‘*Ontology Engineering*’ from ‘*P. Hitzler, M. Krötzsch, and S. Rudolph. Foundations of Semantic Web Technologies. Chapman & Hall/CRC, 2010*’, for download in the TUWEL course.

You can achieve up to a total of 14 points (including the bonus point) for submitting this project.

2 Installation

1. Download the appropriate version of Protégé 5.6.3 and install the program on your computer.¹ Do **not** use earlier versions.
2. NOTE: We only support RDF/XML ontologies. Do not save your project in another format. See the next section for details.

3 Ontology ID and Format

Now create a new ontology by clicking `File` and `New`. Then immediately save the new ontology as

`neighborhoods_XXXXXXXXX.rdf`,

where XXXXXXXXXX are the 8 digits of your student ID. Specifically, click `File` and `Save as`. Select `RDF/XML` as the format.

¹<https://protege.stanford.edu/software.php>

In the initially selected tab `Active Ontology` the ontology's IRI can be configured. Set the following value as ontology IRI:

`http://www.kr.tuwien.ac.at/ontologies/2024S/neighborhoods_XXXXXXXXX.rdf`,
where `XXXXXXXXX` is again your student ID.

Stick strictly to the naming conventions and make sure that you follow the instructions to set the correct preferences in `Protégé 5.6.3` according to the (screenshot) instructions at the end of this project specification.

4 Specification of the TBox and RBox

In the following, a **property** is a statement of **inclusion** between concepts and a **definition** is a statement of **equivalence** between concepts.

Use the following as concept names to specify the ontology:

- `Building`, `Neighborhood`, `Apartment`,
- `ResidentialBuilding`, `CommercialBuilding`, `DualUse`, `Industrial`,
- `Person`, `Tenant`, `Landlord`, `Owner`,
- `LeasedApartment`, `DesirableApartment`,
- `Amenity`, `MetroStation`,
- `School`, `Park`, `Shop`, `Restaurant`,
- `ResidentialNeighborhood`, `FamilyNeighborhood`, `UrbanNeighborhood`,
- `WellLocatedNeighborhood`.

The KB makes use of the following roles:

- `located-in` is *transitive* and its inverse is `contains`.
- `has-amenity` is and its inverse is `is-amenity-of`.
- `owns` and its inverse is `is-owned-by`.
- `rents` and its inverse is `is-rented-by`.
- `lives-in` is a *functional role*, that is to say an entity cannot 'live in' two different entities. Its inverse is `has-resident`.
- `is-near` is *symmetric*, that is to say that if entity *a* is 'near' entity *b*, then *b* is also near *a*.

Translate the following statements into OWL (using the proposed concept and property names) and implement it using `Protégé 5.6.3`.

- (property) A building is located in some neighborhood.
- (definition) There are exactly the following types of buildings: residential, commercial, dual-use and industrial.
- (property) An apartment is located in some residential building or some dual-use building.
- (property) A person lives in some apartment.
- (definition) A tenant is a person that lives in a leased apartment.

- (definition) A leased apartment is an apartment that is rented by at least one person.
- (definition) An owner is a person that owns an apartment.
- (property) Each apartment is owned by at least one and at most four people.
- (definition) A landlord is an owner of a leased apartment.
- (definition) There are exactly the following types of amenities: school, park, shop, restaurant, and metro station.
- (definition) A well-located neighborhood is a neighborhood that has a metro station or is close to another neighborhood that has a metro station.
- (definition) A residential neighborhood is a neighborhood that contains some residential building and no industrial buildings.
- (definition) A family neighborhood is a residential neighborhood that also has (has-amenity) a school and a park.
- (definition) An urban neighborhood is a well-located neighborhood that has (has-amenity role) at least two shops and at least two restaurants.
- (definition) An apartment is desirable if it is in a well-located residential neighborhood.

5 Specification of the ABox

Specify the following ABox assertions and individuals and add them to your KB.

- `Fasanviertel` is a neighborhood that is near `Sonnenwendviertel`.
- `Sonnenwendviertel` is a neighborhood that contains commercial buildings `Capel0` and `Prizeotel`, as well as residential buildings `TheMetropolitan`, `SoVieSo` and `SunQuarter`. `HelmutZilkPark` (park) is also located in this neighborhood, as well as `GTVS-SWQ` school. Furthermore, it has a metro station `Hauptbahnhof`.
- `Mariahilf` is a neighborhood that contains `WeekdayMH`, `TKMaxxMH`, `NikeMH` (shops) and `LeBurger`, `Spear` (restaurants). It has two metro stations `Neubaugasse` and `Zieglergasse`.
- `DonauCity` is a neighborhood that contains `BipaDC`, `PalmersDC` (shops) and `Denito`, `Ikaze` (restaurants). `Donaupark` (park) is also located in this neighborhood. It contains commercial building `AresTower`, `DCTower` and `UNOCity` and residential buildings `DCResidential` and `DanubeFlats`. Metro stations `AlteDonau` and `Kaisermuehlen` are also located in `DonauCity`.
- `Hirschstetten` is a neighborhood that contains the following: `Campanile` (restaurant), `GTVS-Pastinakweg` (school) and `AuPark` (park). Furthermore, it contains the following residential buildings: `Ambiente22`, `House1`, and `House2`.
- `Apt1` is located in `DCResidential`.
- `Apt2` is located in `DanubeFlats`.
- `Apt3` is located in `SoVieSo`.
- `Apt4` is located in `House1`.
- Person `Alice` rents and lives in `Apt1`.

- Person Bob owns Apt2.
- Person Charlie owns and lives in Apt3.
- Person Dave owns Apt1 but lives in Apt4.

Now select `Hermit` as reasoner and execute `Reasoner/Start reasoner`.

6 Analysis Questions

Answer the following questions:

1. Are the conclusions drawn by the reasoner according to your own manually derived conclusions? If not, what could be an explanation? Is the KB consistent? Explain the reasoning.
2. Select 2 axioms from above and write them in DL syntax; also write in DL syntax two inferred axioms (or assertions) and explain the reasoning.
3. Select `Edit/Make all individuals different...` and start the reasoner again. What happens? Is your KB still consistent? What are the results now? Are the conclusions that are drawn by the reasoner now equal to your own conclusions? Explain the reasoning.
4. How does the inferred hierarchy differ from the one you explicitly set up? Are there any residential neighborhoods? If not, what could be the explanation?

You must submit written answers to these questions in the PDF file.

Important: Your answers must be based on the concepts and terminology introduced in the lecture slides.

7 AI Usage Disclosure (Optional Bonus: 1 point)

You may receive **1 additional point** for clearly and honestly stating whether you used any AI tools (e.g., ChatGPT or similar tools) while working on this assignment.

- If you did **not** use AI, explicitly state this.
- If you **did** use AI, provide a brief reflection (3–6 sentences) addressing:
 - What role did AI play in your solution?
 - What limitations or mistakes did you observe?
 - If possible, give one concrete example (i.e., a specific part of your ontology, a DL axiom, or a reasoning result) where AI was helpful and one where it was incorrect or insufficient.

Note: The point is awarded for *honest disclosure and critical reflection*. If you did not use AI, explicitly stating this is sufficient to receive the point, provided the statement is honest.

8 Submission

The deadline for submitting this exercise is **Wednesday, May 20, 23:55**. Submit a ZIP file containing:

- `neighborhoods_XXXXXXX.rdf` - Your OWL ontology file (9 points)
- `answers_XXXXXXX.pdf` - Your answers to the analysis questions (4 points) and the AI usage disclosure (1 point)

where XXXXXXXXX is your student ID, via TUWEL. Before the deadline you can upload your solution several times. Submissions after the deadline will not be accepted and result in no points.

Make sure to strictly stick to the naming conventions (i.e., the concept, property and individual names as proposed, e.g., Neighborhood, Building, Person). Your submitted file has to be stored in the RDF/XML format and the Protégé 5.6.3 preferences have to be set according to the screenshots in Section 10.

IMPORTANT. Submit the latest version of your KB where **all individuals are set to be distinct** but make sure that the KB is consistent!

9 Questions

Please post questions of general interest to the TUWEL forum. Neither complete nor partial solutions must be revealed in the forum. Copying solutions is not allowed. Questions that reveal (part of) your solution should be discussed privately via email to shqiponja.ahmetaj@tuwien.ac.at.

10 Screenshots of the preferences

Please make sure that you have chosen the same properties as shown below within the Protégé 5.6.3 preferences.

The screenshot shows the 'New entities' tab in the Protégé 5.6.3 Preferences dialog. The 'Entity IRI' section has 'Start with' set to 'Active ontology IRI', 'Followed by' set to '#', and 'End with' set to 'User supplied name'. The 'Entity Label (for use with Auto-generated ID)' section has 'Same as label renderer' selected. The 'Auto-generated ID' section has 'Globally unique' selected, with a prefix of 'type_', a suffix of '', a digit count of 20, and a start value of 0. Buttons for 'Reset preferences...', 'OK', and 'Abbrechen' are visible at the bottom.

Preferences

New entities metadata New ontologies OWLViz Plugins Reasoner Renderer User details

Annotations Explanations General Log **New entities**

Entity IRI

Start with: ☒ Active ontology IRI
☐ Specified IRI:

Followed by: ☒ #
☐ /
☐ :

End with: ☒ User supplied name
☐ Auto-generated ID

Entity Label (for use with Auto-generated ID)

☒ Same as label renderer
☐ Custom label

IRI:

Lang:

Auto-generated ID

☐ Numeric (iterative) Prefix:
☒ Globally unique Suffix:
☐ Identitas Digit count:
Start:

The screenshot shows the 'Renderer' tab in the Protégé 5.6.3 Preferences dialog. The 'Entity rendering' section has 'Render by entity IRI short name (Id)' selected. The 'Appearance' section has 'Highlight active ontology statements', 'Show hyperlinks in components that support them', and 'Highlight keywords' all checked. The 'Font size' is set to 14. Buttons for 'Reset preferences...', 'OK', and 'Abbrechen' are visible at the bottom.

Preferences

New entities metadata New ontologies OWLViz Plugins Reasoner **Renderer** User details

Annotations Explanations General Log New entities

Entity rendering ☒ Render by entity IRI short name (Id)
☐ Render by prefixed name
☐ Render by annotation property (e.g., rdfs:label, skos:prefLabel)
☐ Render by prefixed annotation property

Appearance ☒ Highlight active ontology statements
☒ Show hyperlinks in components that support them
☒ Highlight keywords

Font size